VOEDING 12 V

J1 — 12V IN (GND, VIN_RAW)

Q1 — AO4407A (GS_G)

D2 — BZT52C15 (V12 → GS_G)

D1 — SMBJ15CA (V12 → GND)

C4 330µ/25V · C5 330µ/25V · C6 100n/25V · R4 100k · R5 10k

C7 100n

BUCK 12 V –> 5 V 5 A

U1 — TPS54560B (VIN, EN, COMP, RT/CLK, GND, BOOT, SW, FB)

D4 — B560C (V12 → GND)

C8 100n/25V · C9 100n/25V · C10 100n · C11 47µ/25V · R6 100k · R7 24k · R8 10k · R9 68R

C12 47µ/50V · C13 47µ/50V · C14 47µ/50V · R10 5k1 · R11 10k · R12 100k · C15 100n · C16 100n · C17 100n

D5 — groen 5V (GND → LED5_A)

R13 1k

J13 — 5V PI (GND, 5V)

3,3 V (ESP32 + IMU)

D6 — SS14 (VS_ESP → VS)

D7 — SS14 (VS_ESP → VBUS_ESP)

U2 — AMS1117-3.3 (VS_ESP → 3V3)

C18 22µ · C19 100n · C20 10µ

ESP32-S3 (SERVO-BUS + IMU + USB NAAR PI)

U3 — ESP32-S3-WROOM-1-N16R8

D8 — blauw ESP (GND → LED_ESP_A)

R14 10k (ESP_EN) · R15 10k (ESP_IO0) · R16 1k (LED_ESP_A)

C21 100n · C22 100n

R17 10k (BUS_A_RXEN) · R18 10k (BUS_B_RXEN) · R19 10k (IMU_INT)

C23 100n · C24 100n · C25 100n

SW1 — RESET (ESP_EN, GND)

SW2 — BOOT (ESP_IO0, GND)

IMU MPU-6050

U4 — MPU-6050 (VIOSC, VIO, SDA, SCL, ADO, FSYNC, NCS, AUX_DA, AUX_CL, INT, CPOUT, REGOUT, GND)

C26 100n · C27 100n · C28 100n · C29 100n · R20 10k (IMU_SDA) · R21 10k (IMU_SCL)

USB-C NAAR PI 4

J8 — USB-C (VBUS, CC1, CC2, D-, D+, SBU1, SBU2, SHIELD, GND, A1, A4, A5, A6, A7, A8, A9, B5, B6, B7, B8, B9)

USB1 — USB PI

C30 100n · C31 100n

U6 — USBLC6-2SC6 (ESP_DP, ESP_DM)

C32 100n

SENSOR-STROOK (G V S PER GPIO) + I2C

J14 — SENSOR S (G0–G19)

J15 — SENSOR V (V0–V19)

J16 — SENSOR G (GND ×20)

J17 — (GND, 3V3, IMU_SDA, IMU_SCL)

J18 — (GND, 3V3, IMU_SDA, IMU_SCL)

BEVESTIGING

H1 M3 · H2 M3 · H3 M3 · H4 M3

SERVO-BUS A (L1, L2, L3)

U5 — SN74AHCT1G126 (VCC, GND, TX_A, BUSA_DRV)

U8 — SN74LVC1G34 (VCC, GND, BUSA_RXIN, RX_A)

R22 10k (BUSA_DRV) · R23 10k (BUSA_RXIN) · R24 10k (RX_A) · R25 10k (BUS_A_RXEN) · C33 100n · C34 100n

J2 — L1 servo (GND, 12V, BUS_A)

J3 — L2 servo (GND, 12V, BUS_A)

J4 — L3 servo (GND, 12V, BUS_A)

SERVO-BUS B (L4, L5, L6)

U7 — SN74AHCT1G126 (VCC, GND, TX_B, BUSB_DRV)

U9 — SN74LVC1G34 (VCC, GND, BUSB_RXIN, RX_B)

R26 10k (BUSB_DRV) · R27 10k (BUSB_RXIN) · R28 10k (RX_B) · R29 10k (BUS_B_RXEN) · C35 100n · C36 100n

J5 — L4 servo (GND, 12V, BUS_B)

J6 — L5 servo (GND, 12V, BUS_B)

J7 — L6 servo (GND, 12V, BUS_B)

ARACHNE-18 MINI BRAIN v1.0 – ESP32-S3 + MPU-6050 hexapod servo controller for a Raspberry Pi 4 (generated by tools/build_sch.py from tools/design.py)
Servo connectors: 1 = GND, 2 = +12 V, 3 = DATA (Feetech STS half-duplex bus, 1 Mbaud). CHECK the pin order against your cable before plugging in!
Bus A = legs L1 L2 L3 (ESP32 UART1: TX IO17, RX IO18, DE IO16). Bus B = legs L4 L5 L6 (UART2: TX IO10, RX IO11, DE IO9). DE = RTS in RS485 half-duplex mode.
Pi 4 link: USB-C (board) <-> USB-A (Pi) cable on the ESP32-S3 native USB -> /dev/ttyACM0 on the Pi. Pi VBUS only feeds the ESP32 (through D7), never the servos.
5V PI pads: buck output (5 V, 3 A for the Pi 4) -> Pi GPIO pin 2/4 (+5V) and 6 (GND), or a cut USB-C lead into the Pi power socket.
Buck = TI TPS54560B datasheet design (fig. 32): 400 kHz, 5.0 V 5 A, UVLO start 6.5 V.

ESP32-S3 + MPU-6050, 6 servo-bus legs, USB to Pi 4
Kobe Van Robaeys – maakleerplek
Sheet: /
File: arachne_mini.kicad_sch
Title: ARACHNE-18 mini brain board
Size: A1 | Date: 2026-10-18 | Rev: 1.0
KiCad E.D.A. 10.0.3 | Id: 1/1